Function Define

Absolute Function:

$$f(x) = \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$$

1. $f(x) = \begin{cases} |x| + 1 & ; \quad x > 0 \\ 1 & ; \quad x = 0 \\ |x| + 2 & ; \quad x < 0 \end{cases} = \begin{cases} x + 1 & ; \quad x > 0 \\ 1 & ; \quad x = 0 \\ -x + 2 & ; \quad x < 0 \end{cases}$

As $x > 0$, then $f(x) = |x| + 1 = x + 1$

As $x < 0$, then $f(x) = |x| + 2 = -x + 2$

2. $f(x) = |x| - 3 = \begin{cases} x - 3 & ; \quad x \geq 0 \\ -x - 3 & ; \quad x < 0 \end{cases}$

3. $f(x) = \frac{|x|}{x} = \begin{cases} \frac{x}{x} & ; \quad x > 0 \\ \frac{-x}{x} & ; \quad x < 0 \end{cases}$

4. $f(x) = \frac{x}{|x|} = \begin{cases} \frac{x}{x} & ; \quad x > 0 \\ \frac{x}{-x} & ; \quad x < 0 \end{cases}$

As here, if $x = 0$, then $\frac{|x|}{x}$ or $\frac{x}{|x|}$ is undefined. So equal to 0 case is not considered here.

$[\frac{1}{0}, \frac{2}{0} \ldots = \text{Undefined}]$

5. $f(x) = \begin{cases} \frac{|x-3|}{x-3} & ; \quad x \neq 0 \\ 0 & ; \quad x = 3 \end{cases} = \begin{cases} \frac{-(x-3)}{(x-3)} & ; \quad x - 3 < 0 \\ 0 & ; \quad x = 3 \\ \frac{(x-3)}{(x-3)} & ; \quad x - 3 > 0 \end{cases}$

$$= \begin{cases} \frac{-(x-3)}{(x-3)} & ; \quad x < 3 \\ 0 & ; \quad x = 3 \\ \frac{(x-3)}{(x-3)} & ; \quad x > 3 \end{cases}$$

6. $f(x) = |x-3| = \begin{cases} (x-3) & ; \quad (x-3) \geq 0 \\ -(x-3)\,; & \quad (x-3) < 0 \end{cases}$

$$= \begin{cases} (x-3) & ; \quad x \geq 3 \\ -(x-3)\,; & \quad x < 3 \end{cases}$$

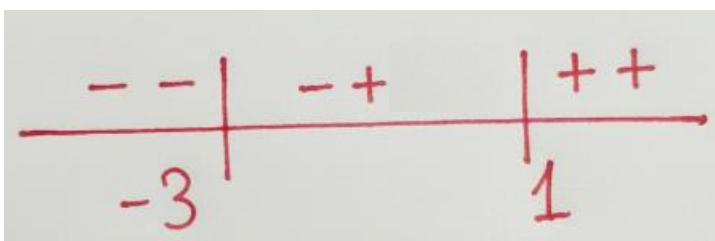
7. $f(x) = \frac{|x-3|}{x-3} = \begin{cases} \frac{-(x-3)}{(x-3)} & ; \quad (x-3) < 0 \\ \frac{(x-3)}{(x-3)} & ; \quad (x-3) > 0 \end{cases} = \begin{cases} \frac{-(x-3)}{(x-3)} & ; \quad x < 3 \\ \frac{(x-3)}{(x-3)} & ; \quad x > 3 \end{cases}$

Here, for $f(x) = \frac{|x-3|}{x-3}$, $(x-3) \neq 0 \Longrightarrow x \neq 3$

As, if $(x-3) = 0 \Longrightarrow x = 3$, then, $\frac{|x-3|}{x-3}$ = Undefined as $\frac{1}{0}, \frac{2}{0}, \frac{3}{0} \ldots$ = Undefined. So equal to 3 case is not considered here.

8. $f(x) = |x-1| + |x+3|$

If $x - 1 = 0 \Longrightarrow x = 1$ and $x + 3 = 0 \Longrightarrow x = -3$



If $x \geq 1$,

Let $x = 2$, then,

$x - 1 = 2 - 1 = 1\ (+ve)$

$x + 3 = 2 + 3 = 5\ (+ve)$

If $-3 \leq x < 1$,

Let $x = -2$, then,

$x - 1 = -2 - 1 = -3\ (-ve)$

$x + 3 = -2 + 3 = 1\ (+ve)$

If $x < -3$,

Let $x = -4$, then,

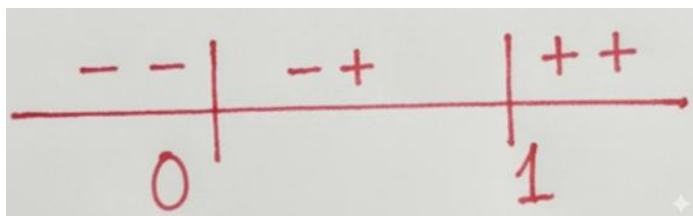
$x - 1 = -4 - 1 = -5\ (-ve)$

$x + 3 = -4 + 3 = -1\ (-ve)$

So, $f(x) = \begin{cases} (x-1)+(x+3) & ; \quad x \geq 1 \\ -(x-1)+(x+3)\,; & -3 \leq x < 1 \\ -(x-1)-(x+3)\,; & x < -3 \end{cases} = \begin{cases} 2x+2 & ; \quad x \geq 1 \\ 4 & ; \quad -3 \leq x < 1 \\ -2x-2 & ; \quad x < -3 \end{cases}$

9. $f(x) = |x-1| + |x+0| = |x-1| + |x|$

If $x - 1 = 0 \Longrightarrow x = 1$ and $x = 0$.



If $x \geq 1$,

Let $x = 2$, then,

$x - 1 = 2 - 1 = 1\ (+ve)$

$x = 2\ (+ve)$

If $0 \leq x < 1$,

Let $x = 0$, then,

$x - 1 = 0 - 1 = -1\ (-ve)$

$x = 0\ (+ve)$

If $x < 0$,

Let $x = -2$, then,

$x - 1 = -2 - 1 = -3\ (-ve)$

$x = -2\ (-ve)$

So, $f(x) = \begin{cases} (x-1)+x & ; \quad x \geq 1 \\ -(x-1)+x & ; \quad 0 \leq x < 1 \\ -(x-1)-x & ; \quad x < 0 \end{cases} = \begin{cases} 2x-1 & ; \quad x \geq 1 \\ 1 & ; \; 0 \leq x < 1 \\ -2x+1 & ; \quad x < 0 \end{cases}$

10. $f(x) = \begin{cases} \sqrt{|x|}; & x \geq 0 \\ -\sqrt{|x|}; & x < 0 \end{cases} = \begin{cases} \sqrt{x}; & x \geq 0 \\ -\sqrt{-x}; & x < 0 \end{cases}$

Here, for $x < 0$, the value of x is a negative number. So, here we take $\sqrt{-x}$. By taking $\sqrt{-x}$ we are taking the negative of a negative number which makes the number positive. So we actually have a positive number inside the square root.